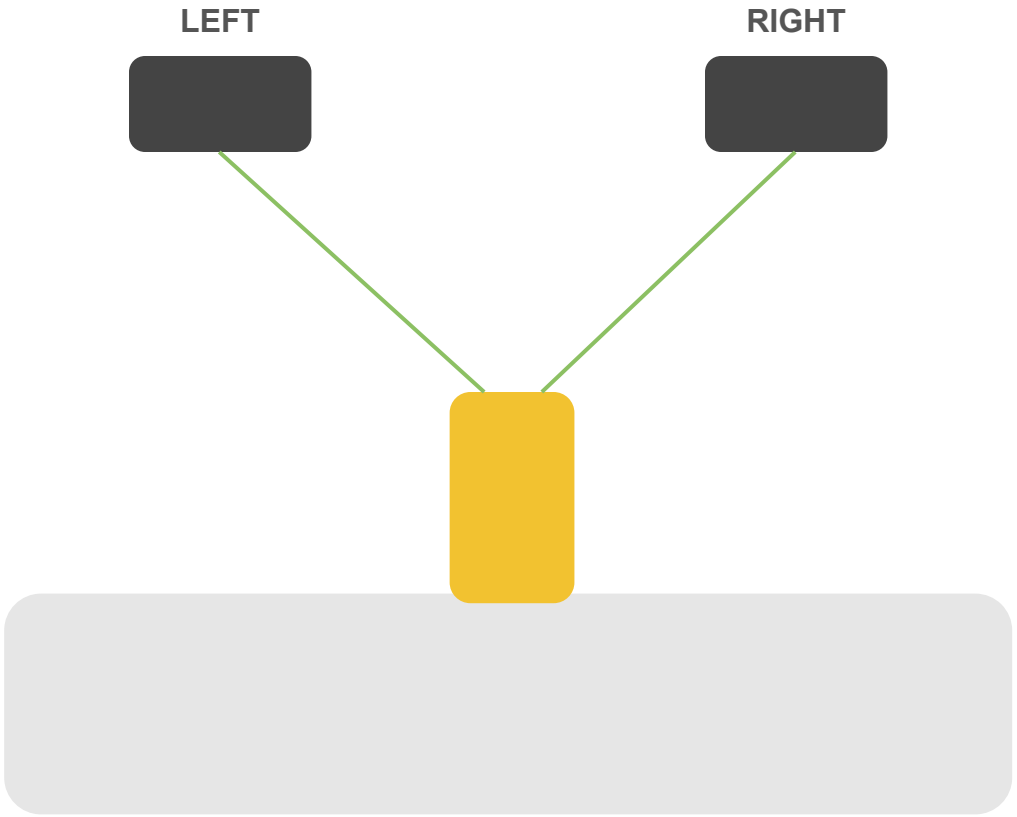
ROBOTIC ARM COURSE · LESSON 10 · STEREO VISION

Stereoscopic Camera

Measuring distance with two cameras, and calibrating them so the numbers can be trusted

Day 1 · Depth from disparity

Day 2 · Camera calibration and 3D points for the arm



WHY STEREO

One Camera Cannot Measure Distance

Last lesson, one detection told the arm which direction to turn. It could not say how far to reach.

SMALL AND NEAR
A small cup close to the camera

BIG AND FAR
A large cup far from the camera

SAME PICTURE
Both can fill the same pixels, so one image cannot tell them apart.

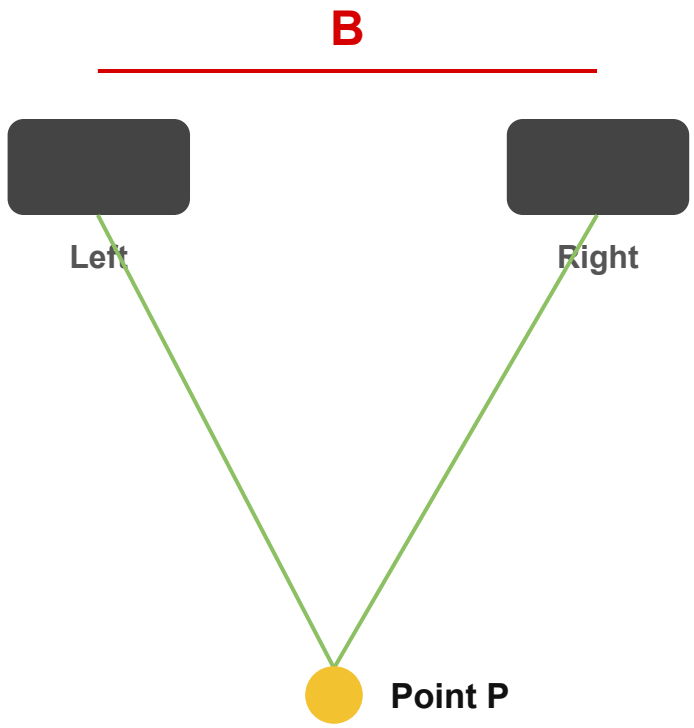
IDEA
Two cameras a known distance apart give two views. The difference between the views tells us the distance.

TRY IT
Hold up a finger, close one eye, then swap eyes. The finger jumps, but the far wall barely moves.

DAY 1 · TWO VIEWS

Two Cameras, Two Views

Place two identical cameras side by side. The distance between them is the **baseline B**.



WHAT CHANGES

A nearby point lands at different x positions in the left and right images.

WHAT STAYS THE SAME

Its y position, because the cameras sit at the same height and look the same way.

DAY 1 · DISPARITY

Disparity: How Far the Point Moved

Find the same point in both images. The shift in x is the disparity.

DEFINITION

$$d = x_L - x_R$$

Measured in pixels. Always positive for a point in front of the cameras.

BIG DISPARITY

The point is **close**. The two views disagree a lot.

SMALL DISPARITY

The point is **far**. The two views almost agree.

Point	x_L	x_R	d
Near cup	420	350	70
Far box	410	390	20

DAY 1 · THE FORMULA

Depth from Disparity

FORMULA

$$Z = f \cdot B \div d$$

Z: depth in mm. f: focal length in pixels. B: baseline in mm. d: disparity in pixels.

REARRANGED

$$d = f \cdot B \div Z$$

The same formula solved for disparity. Handy for checking.

UNITS

f and d are both in pixels, so the pixels cancel and Z comes out in the units of B.

SANITY CHECK

Disparity doubles when depth halves. Close things move more between views.

COMMON MISTAKE

Mixing units: B in cm gives Z in cm. Keep B and Z in mm.

DAY 1 · WORKED EXAMPLE

Depth Example

f = **700 px**, B = **60 mm**

The cup appears at x_L = **420** and x_R = **350**.

How far away is the cup?

SOLUTION

d = 420 − 350 = 70 px

Z = 700 × 60 ÷ 70 = 42,000 ÷ 70

Z = 600 mm

DAY 1 · TRY IT YOURSELF

Try It Yourself #1

f = **800 px**, B = **50 mm**

x_L = **500**, x_R = **420**

Find the disparity and the depth.

SOLUTION

d = 500 − 420 = 80 px

Z = 800 × 50 ÷ 80 = 500 mm

DAY 1 · TRY IT YOURSELF

Try It Yourself #2

Going the other way: how much disparity should we expect?

f = **800 px**, B = **50 mm**

An object sits at Z = **250 mm**.

What disparity should the matcher find?

SOLUTION

$d = f \cdot B \div Z = 800 \times 50 \div 250 = 40{,}000 \div 250$

$d = 160$ px

DAY 1 · ACCURACY

Far Away Means Less Accurate

Matching is only good to about 1 pixel. Here is what a 1 px mistake does with f = 700 px and B = 60 mm.

True depth	True d	Measured d	Measured Z	Error
500 mm	84 px	83 px	506 mm	6 mm
1000 mm	42 px	41 px	1024 mm	24 mm
2000 mm	21 px	20 px	2100 mm	100 mm

TAKEAWAY

Error grows roughly with the **square** of the distance. Doubling the distance makes the same 1 px mistake about four times worse.

DAY 1 · DESIGN CHOICE

Choosing the Baseline

LARGER B

Better far accuracy

More disparity, so a 1 px mistake matters less. But the two views overlap less, and very close objects are lost from one view.

SMALLER B

Better near coverage

Views overlap well and close objects are visible. But disparity is small, so far depths get noisy.

FOR OUR ARM

The arm works within 500 mm, so a baseline of a few centimeters is enough. The arm's RealSense D435i camera has **B = 50 mm**.

DAY 1 · CHECK WITH CODE

Check It with Code

Place the object on a tape mark. Read the column of its center in the left and right images.

PYTHON

```
from stereo_helper import focal_px, depth_from_disparity

f = focal_px(640, 60)          # Lesson 9: about 554 px
x_L, x_R = 412, 298            # read from the two images
Z = depth_from_disparity(x_L - x_R, f, 60)
print(round(Z), "mm")          # 292
```

YOUR JOB

Repeat on the **300**, **600** and **1200 mm** marks. Compare each answer with the tape.

EXPECT

The error grows at 1200 mm. Why? f is only a data sheet estimate until Day 2.

DAY 1 · RECAP

Day 1 Recap

DISPARITY

Pixel shift

$d = x_L - x_R$

DEPTH

Near = big d

$Z = f \cdot B \div d$

ERROR

Grows with Z^2

1 px of error costs more at long range

BASELINE

B matters

Trade accuracy against overlap

COMING UP

Next time: where do f and the other camera numbers come from? We calibrate the cameras.

DAY 2

Calibration and 3D Points

Day 1 assumed we knew f and that the images were perfectly aligned. Now we measure them for real and turn pixels into positions the arm can reach.

DAY 1 IN ONE LINE

$Z = f \cdot B \div d$, and 1 px of error costs more far away

DAY 2 · CAMERA NUMBERS

What Calibration Finds

INTRINSICS K

Inside each camera

Focal lengths fx, fy and center (cx, cy), all in pixels. Different for every camera.

DISTORTION

Lens bending

How the lens bends straight lines. We use it to straighten the image.

R AND T

Between cameras

How the right camera is rotated and shifted from the left. The length of T is the baseline B.

INTRINSIC MATRIX K

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

DAY 2 · COLLECTING IMAGES

The Checkerboard

BOARD

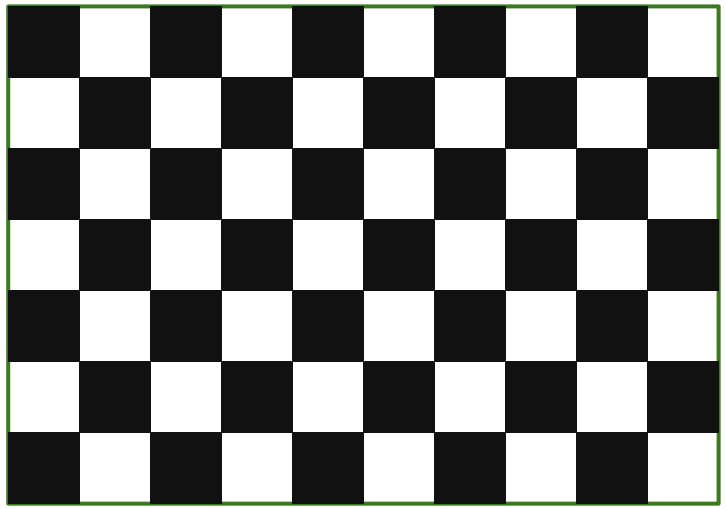
10 × 7 squares, so 9 × 6 inner corners. 25 mm squares, measured with a ruler.

PHOTOS

15 to 20 pairs. Tilt it, move it near and far, cover the whole frame. Both cameras must see the whole board.

WHY IT WORKS

The software knows where each corner really is on the board, and sees where it lands in the image.



9 × 6 inner corners
where four squares meet

DAY 2 · CHECKING THE RESULT

Reprojection Error

Use the new numbers to predict where each corner should appear, then compare with where it was found.

GOOD

Under 0.5 px

Depth from this camera pair can be trusted.

POOR

Over 1 px

Redo it. Remove blurry pairs and take more varied views.

COMMON CAUSES OF A POOR RESULT

Board not flat, motion blur, board only at one angle, board partly out of frame, wrong square size, cameras moved.

DAY 2 · RECTIFICATION

Making the Rows Line Up

Our formula assumes a matching point is on the **same row** in both images. Rectification warps both images so that is true.

BEFORE

A point may sit on row 200 in the left image and row 207 in the right. The matcher would have to search everywhere.

AFTER

Both images are straightened and aligned. The match is on the same row, so the search is fast.

WATCH OUT

Rectifying changes the focal length and center. Always use the new f, cx and cy it returns in the depth formulas.

DAY 2 · DISPARITY MAP

A Disparity for Every Pixel

StereoSGBM compares small patches along each row and gives a disparity map. Brighter pixels mean closer.

MAX_DISP

e.g. 160

Largest disparity it searches. Must be a multiple of 16.

CLOSEST MEASURABLE DEPTH

700 × 60 ÷ 160 ≈ 263 mm

Z_min = f · B ÷ max_disp

WHY IT MATTERS

An object closer than Z_min needs a disparity larger than the search range, so the matcher cannot find it. The arm works from about 300 to 500 mm, so 160 covers it.

DAY 2 · TRY IT YOURSELF

Try It Yourself #3

f = **700 px**, B = **60 mm**

max_disp = **64**

What is the closest depth that can be measured?

SOLUTION

Z_min = f · B ÷ max_disp = 42,000 ÷ 64

Z_min ≈ 656 mm

The arm’s 300 to 500 mm range would be lost. Use 160 to reach down to about 263 mm.

DAY 2 · PIXEL TO 3D

A Pixel Becomes a 3D Point

With depth Z known, the pixel's position gives the sideways and vertical offsets, in the camera's frame.

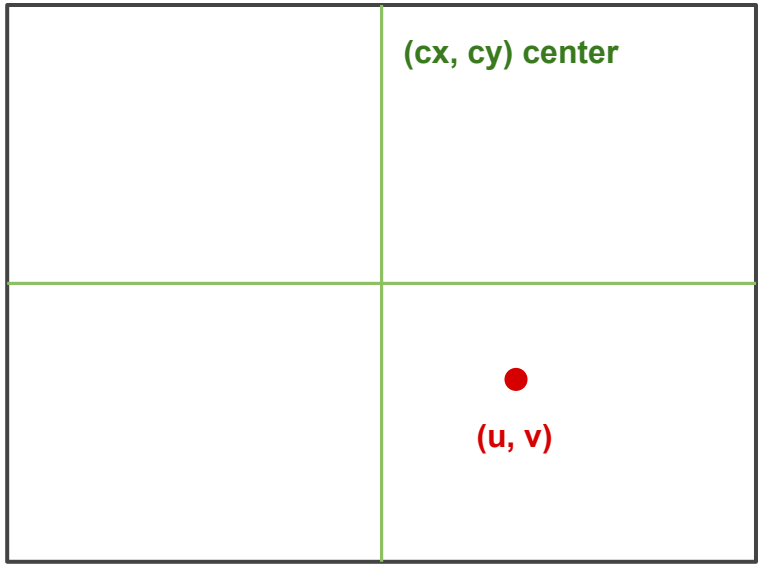
CAMERA FRAME

$$X = (u - cx) \cdot Z \div f$$

$$Y = (v - cy) \cdot Z \div f$$

$$Z = f \cdot B \div d$$

Same top-left pixel origin as Lesson 9: v grows downward, so Y points down.



X is right, Y is down, Z is forward.

DAY 2 · WORKED EXAMPLE

Pixel to Camera Frame

f = **700 px**, cx = **320**, cy = **240**

Pixel (u, v) = **(400, 300)**, depth Z = **600 mm**

SOLUTION

X = (400 − 320) × 600 ÷ 700 = 68.6 mm

Y = (300 − 240) × 600 ÷ 700 = 51.4 mm

Point = (68.6, 51.4, 600) mm

DAY 2 · TRY IT YOURSELF

Try It Yourself #4

f = **800 px**, cx = **320**, cy = **240**

Pixel (u, v) = **(240, 300)**, depth Z = **400 mm**

Find X and Y.

SOLUTION

X = (240 − 320) × 400 ÷ 800 = −40 mm

Y = (300 − 240) × 400 ÷ 800 = 30 mm

Point = (−40, 30, 400) mm

Negative X means left of the camera’s center line.

DAY 2 · INTO THE ARM'S WORLD

Camera Frame to Base Frame

The camera sits at (cam_x, cam_y, cam_z) in the base frame and looks along the arm's +x direction.

CONVERSION

$$x_b = cam_x + Z$$
$$y_b = cam_y - X$$
$$z_b = cam_z - Y$$

WHY THE SIGNS FLIP

Camera X points right, but the arm's +y points left. Camera Y points down, but +z points up.

THEN CHECK REACH

The arm has two 250 mm links, so a target farther than 500 mm from the shoulder is out of reach.

DAY 2 · WORKED EXAMPLE

Where Should the Gripper Go?

Camera at **(100, 0, 150) mm** in the base frame.

Camera-frame point from the cup: **(34.3, 25.7, 300) mm**

SOLUTION

$x_b = 100 + 300 = 400$ mm

$y_b = 0 - 34.3 = -34.3$ mm

$z_b = 150 - 25.7 = 124.3$ mm

Distance = $\sqrt{(400^2 + 34.3^2 + 124.3^2)} = 420.3$ mm

Target (400, −34.3, 124.3), within reach

$\theta_B = \text{atan2}(-34.3, 400) = -4.9°$

WHY NOT THE LESSON 9 ANGLE?

From the camera, the cup is atan(34.3 ÷ 300) = **6.5°** to the right. The base sits 100 mm behind the camera, so it only needs to turn **4.9°**.

DAY 2 · CHECK WITH CODE

The Whole Workflow in Code

PYTHON

```
from stereo_helper import *

rms1, K1, d1 = calibrate_camera(obj, pts_L, size)
rms2, K2, d2 = calibrate_camera(obj, pts_R, size)
rms, R, T = calibrate_stereo(obj, pts_L, pts_R, K1, d1, K2, d2, size)
L, Rt, f, cx, cy = rectify_pair(left, right, K1, d1, K2, d2, R, T, size)
disp = disparity_map(L, Rt, max_disp=160)
Z = depth_from_disparity(median_disparity(disp, box), f, baseline_mm(T))
u, v = (box[0] + box[2]) / 2, (box[1] + box[3]) / 2   # box from Lesson 9
P = pixel_to_camera(u, v, Z, f, cx, cy)
target = camera_to_base(P, cam_pos=(100, 0, 150))
```

CHECK

Use the worked examples. You should get (68.6, 51.4, 600) and (400, −34.3, 124.3).

ALWAYS

Print the reprojection error first. Do not trust depth if it is above 1 px.

WRAP-UP

What We Can Now Do

DAY 1

Turn disparity into depth and judge how accurate it is

DAY 1

Choose a baseline that fits the working range

DAY 2

Calibrate a stereo pair and check reprojection error

DAY 2

Rectify, build a disparity map and turn a pixel into arm coordinates

PUTTING IT TOGETHER

Detection tells the arm what and which way. Stereo tells it how far. Together they give a 3D target, and inverse kinematics from Lessons 6 to 8 turns it into joint angles.